

Company Name: D. E. Shaw & Co.

Role: Systems Intern

Candidate Name: Debabrata Mondal

Online Assessment :

Given an integer and two arrays A ($A_1, \dots, A_n$) and B ($B_1, B_2, \dots, B_n$) you are allowed to perform the following two operations on it –

1. Choose an index i in the array and assign $A_i = A_i + B_i$.
2. Delete one element from either the beginning or the end of both the arrays simultaneously.

You are only allowed to perform operation (1) a maximum of number of times.

Determine the maximum possible sum of array A after performing any number of operations in total.

=> Kadane's algorithm for finding the maximum subarray sum is helpful for solving this.

The MCQs mostly cover CS fundamentals (DBMS and OS topics are important)

Technical Interview 1 (08/08/2023)

Initially, we began discussing my solution to the OA DSA question. The interviewers then went through my resume as well as my GitHub account and asked questions from there.

I had a project that used Redis and they asked how I had deployed that. They wanted to understand the architectural choices behind the project. Why Redis? Why do you need to cache things? How do you invalidate caches?

Another open source project of mine piqued their interest. It was about a vector database. They wanted to know why I created that, how it is different from other forms of databases. I shared my reasonings and also the journey that went into creating it – going to Dubai, speaking with investors, hackathons, etc.

Finally they asked about the implementation of Git. If I had to implement it, what would I have done? How can version tracking be improved? What's the need for a version control system?

HR Round (08/08/2023)

The HR round involved conversations around my interests outside of university and studies. Since I was a part of the ACM Student Chapter, I was asked questions around my role, the club's activities, etc. The interviewer also asked why I wanted to work at D. E. Shaw and what interested me about the firm.

Technical Interview 2 (09/08/2023)

The interview began with a discussion on ML, LLMs and the transformers architecture. None of these were mentioned in my resume, I just happened to tell them I took an interest in that field.

The interviewers then proceeded to ask me about my experience with React.js. They wanted to know whether I knew how React works internally, so the questions were surrounding advanced React concepts.

We then started discussing computer networks and the cloud. They wanted to know that if I had the opportunity, how would I design my own private cloud. Understanding of IP addresses and their various types were helpful.

I had a project in my resume which used FastAPI. The interviewers wanted to know why I used it, what alternatives had I looked into, etc.

The same project also used Postgres DB. They asked me about the different types of databases, ACID properties, etc. Finally, the interviewers asked me about database indices, how they worked, the pros and cons of different types of indices.

The interview ended with a DSA question: Leetcode medium level of difficulty which involved some pointer arithmetic!

Verdict: Selected